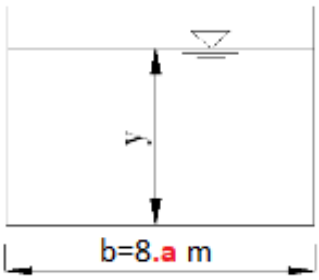
Date:13.01.2021

2020-2021 FALL SEMESTER HYDRAULICS COURSE FINAL EXAM

Name: **Surname:** **Student ID:**

Question 3 (25):



A rectangular **wide channel** with b=8.**a** m, n=0.014 (Manning coefficient), and J_0=0.0001 (bottom slope) carries a discharge of 7**b** m^3/s. A dam built across the channel and the height of the water just upstream of the dam raises to 10m.

a) Determine the water heights at a distance of Δx=10000m upstream of the dam using Bakhmeteff Method.

b) Draw the water surface profile.

a is the second last digit of the student number
b is the last digitr of the student number
Student number 17042082 a=8 b=2